Chapter Five

Digital electronics and IC fabrication techniques

Brief overview of Basic and Derived logic gates

Circuits which are used to process digital signals are called logic gates. They are binary in nature. Gate is a digital circuit with one or more inputs but with only one output. The output appears only for certain combination of input logic levels. Logic gates are the basic building blocks from which most of the digital systems are built up. The numbers 0 and 1 represent the two possible states of a logic circuit. The two states can also be referred to as 'ON and OFF' or 'HIGH and LOW' or 'TRUE and FALSE'.

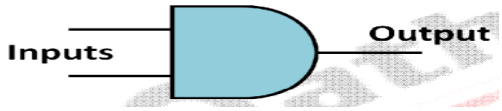
Basic logic gates using discrete components

The basic elements that make up a digital system are 'OR', 'AND' and 'NOT' gates. These three gates are called basic logic gates. All the possible inputs and outputs of a logic circuit are represented in a table called TRUTH TABLE. The function of the basic gates are explained below with circuits and truth tables.

AND gate

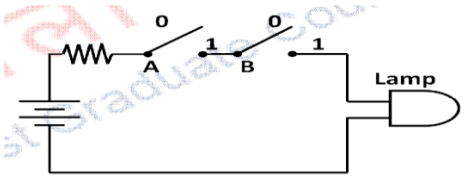
The AND gate is an electronic circuit that gives a **high** output (1) only if **all** its inputs are high. A dot (.) is used to show the AND operation i.e. A.B. Bear in mind that this dot is sometimes omitted i.e. AB

The logical operation of AND gate can be expressed with the help of a table which includes all input combinations and corresponding outputs. Such table of input / output relations is known as truth table. In this case, there are 2 inputs to AND gates to provide $2^2 = 4$ input combinations in the truth table. The truth table may be extended to any number of inputs. For 3 inputs, there are 8 combinations and 16 for 4 inputs.



Input A (Switch)	Input B (Switch)	Output Y (Lamp)
0 (Open)	0 (Open)	0 (OFF)
0 (Open)	1 (Closed)	0 (OFF)
1 (Closed)	0 (Open)	0 (OFF)
1 (Closed)	1 (Closed)	1 (ON)

The AND gate produces a logic 1, if and only if all the inputs are at logic 1.

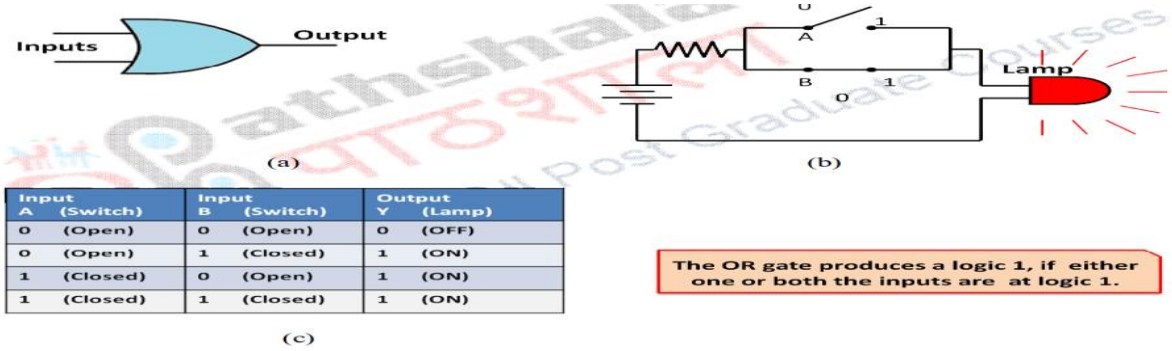


Switch implementation

OR Gate

An OR gate is a logic circuit with two or more inputs and one output. The OR gate performs logical addition i.e. OR function. Fig. 3 shows the logic symbol and truth table of two input OR gate along with simple implementation using switches. The output of an OR gate is HIGH only when all of its inputs are in the HIGH state. In all other cases, the output is LOW.

For OR gate, **Y = A + B**



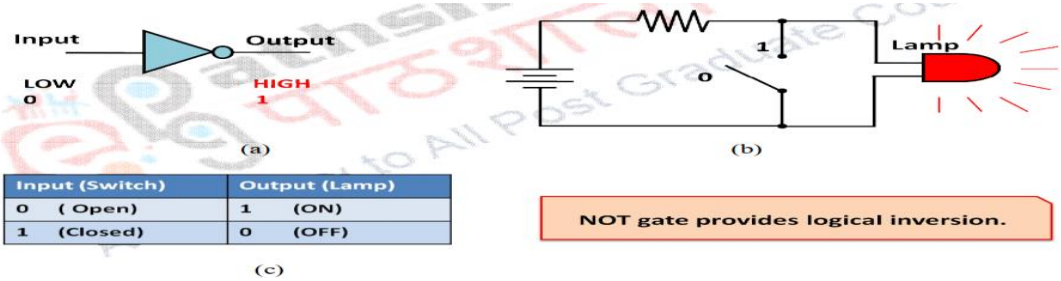
Input A (Switch)		Input B (Switch)		Output Y (Lamp)	
0	(Open)	0	(Open)	0	(OFF)
0	(Open)	1	(Closed)	1	(ON)
1	(Closed)	0	(Open)	1	(ON)
1	(Closed)	1	(Closed)	1	(ON)

(c)

Figure 3: OR gate (a) Logic Symbol (b) Switch implementation (c) Truth table.

NOT gate:

NOT gate has one-input and one-output. It is a logic circuit whose output is always the complement of the input. Fig. 1 indicates the logic symbol and truth table of NOT gate along with simple implementation using switches. The NOT gate is popularly known as **inverter**. It performs logical inversion or complementation. That is, if A is the input to a NOT circuit, then its output Y is given by **Y =** $\overline{A}$ **or A'**. That is, a LOW input produces a HIGH output and vice versa.



Input (Switch)		Output (Lamp)	
0	(Open)	1	(ON)
1	(Closed)	0	(OFF)

(c)

Figure 1: NOT gate (a) Logic Symbol (b) Switch implementation (c) Truth table.

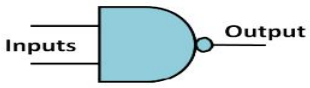
NAND gate

NAND gate is combination of AND and NOT gates. The NAND gate provides AND functions with inverted output. Fig.4 indicates the logic symbol and truth table of two input NAND gate along with simple implementation using switches. The output of a NAND gate is a logic '0' when all its inputs are a logic '1'. For all other input combinations, the output is a logic '1'. NAND gate operation is logically expressed as $Y = \overline{A.B}$

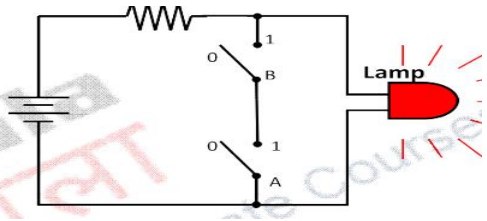
Input A (Switch)	Input B (Switch)	Output Y (Lamp)
0 (Open)	0 (Open)	1 (ON)
0 (Open)	1 (Closed)	1 (ON)
1 (Closed)	0 (Open)	1 (ON)
1 (Closed)	1 (Closed)	0 (OFF)

The NAND gate produces a logic 1, if either one or both the inputs are at logic 0 else the output is at logic 1.

Truth table



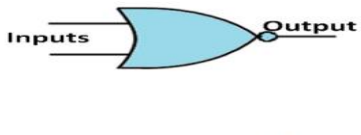
Logic Symbol (b)



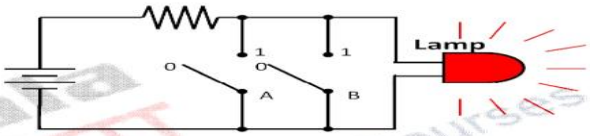
Switch implementation (c)

NOR gate

NOR gate is combination of OR and NOT gates. The NOR gate provides OR function with inverted output. Fig. 5 shows the logic symbol and truth table of two input NOR gate along with simple implementation using switches. The output of a NOR gate is a logic '1' when all its inputs are logic '0'. For all other input combinations, the output is a logic '0'. NOR gate operation is logically expressed as $Y = A + B$



(a)



(b)

Input A (Switch)	Input B (Switch)	Output Y (Lamp)
0 (Open)	0 (Open)	1 (ON)
0 (Open)	1 (Closed)	0 (OFF)
1 (Closed)	0 (Open)	0 (OFF)
1 (Closed)	1 (Closed)	0 (OFF)

(c)

The NOR gate produces a logic 0, if either one or both the inputs are at logic 1 else the output is at logic 0.

Figure 5: NOR gate (a) Logic Symbol (b) Switch implementation (c) Truth table.

XOR (Exclusive OR) gate

Exclusive OR gate is basically designed to exclude the condition of standard OR gate so as to generate real binary addition. An XOR gate is a two inputs and one output logic circuit. Fig.6 indicates the logic symbol and truth table of two input XOR gate along with simple implementation using switches. The output of an XOR gate is at logic '1' when the inputs are dissimilar and at logic '0' when the inputs are similar. The logic equation for two input XOR gate is given by $Y = A \oplus B$

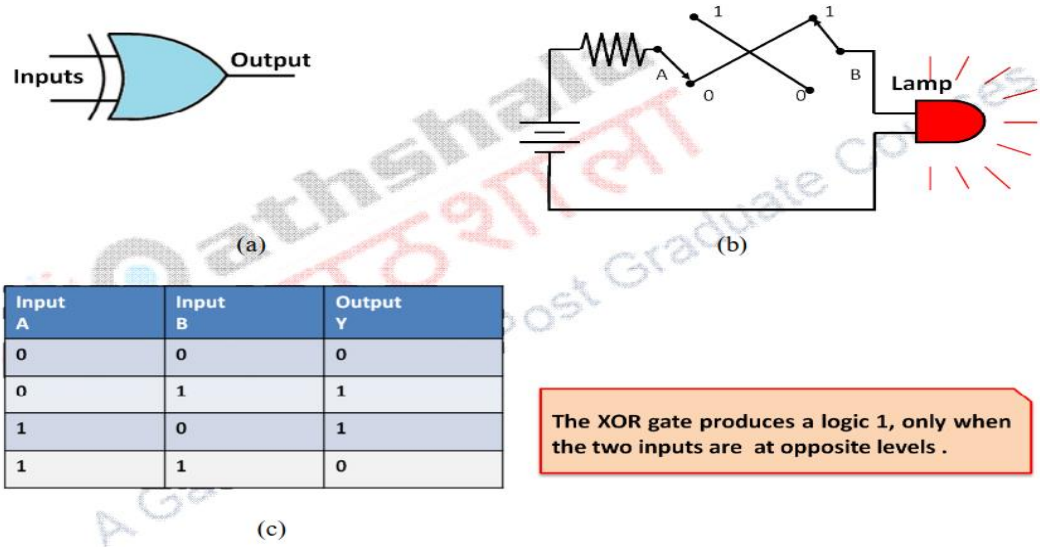


Figure 6: XOR gate (a) Logic Symbol (b) Switch implementation (c) Truth table.

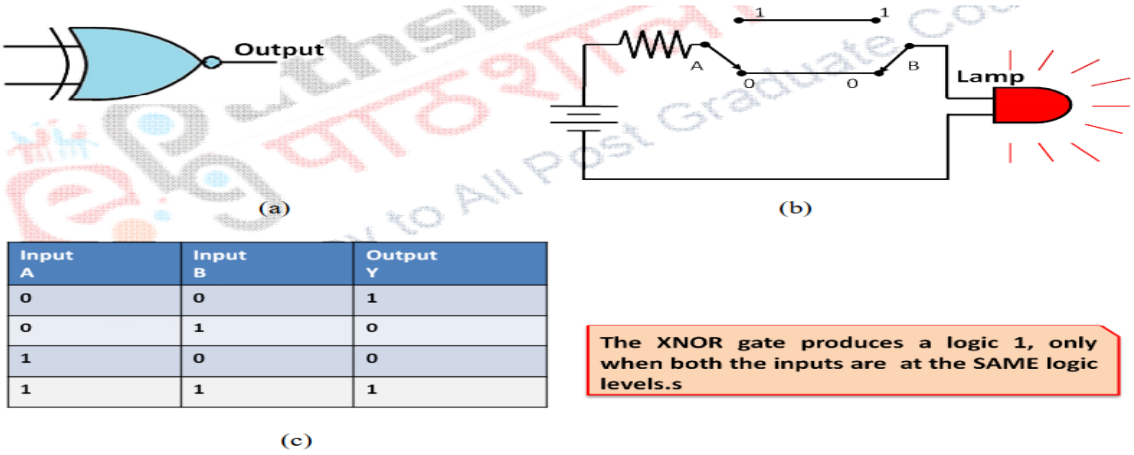
When more than two inputs are to be XORed then multiple two inputs XOR gates may be used. The output of a multiple-input XOR logic function is at logic '1' when the number of 1s in the input sequence is odd and at logic '0' when the number of 1s in the input sequence is even, including zero. That is, an all 0s input sequence also produces a logic '0' at the output. Three or more inputs XOR do not exist.

XNOR gate

XNOR is obtained by the combination of NOT and XOR gates. XNOR gate is a two inputs and one output XOR gate with active low output. Fig.7 indicates the logic symbol and truth table of two input XNOR gate along with simple implementation using switches. The output of an XNOR gate is at logic '1' when the inputs are similar and at logic '0' when the inputs are dissimilar. The logic equation for two input XOR gate is given by

$$\mathbf{Y} = \overline{A\overline{B} + \overline{A}B}$$

The output of XNOR can be obtained by complementing the output column for XOR gate for corresponding input conditions.



Input A	Input B	Output Y
0	0	1
0	1	0
1	0	0
1	1	1

(c)

The XNOR gate produces a logic 1, only when both the inputs are at the SAME logic levels.s

Figure 7: XNOR gate (a) Logic Symbol (b) Switch implementation (c) Truth table.

Universal Logic Gates

Any Boolean / logic expression can be realized using the AND, OR, and NOT gates. From these three primary gates, two derived gates NAND and NOR are usually realized. It is possible to construct basic gates namely NOT, AND, OR using combination of NAND gates or a combination of NOR gates. For this reason NAND and NOR gates are called as universal logic gates.

Let us now understand the importance of NAND and NOR logic gates. The NAND and NOR gates are referred to as Universal logic gates as each of these gates can be used to construct basic logic gates AND, OR & NOT gates.

NAND gate as universal logic gate

Let us consider now NAND gate as a universal logic gate. Figure 9 shows the implementation of basic logic gates using only NAND gates.

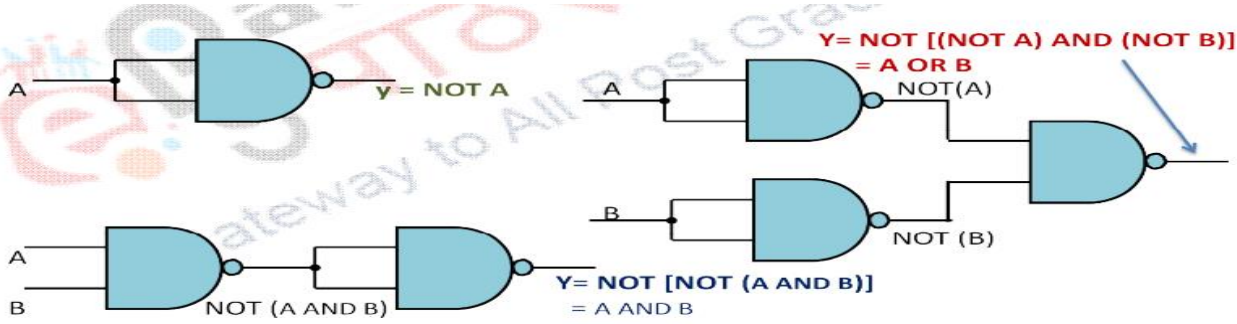


Figure 9: Construction of NOT, AND & OR gates using NAND gates

NAND gate can be used to as NOT gate by connecting all its inputs together and applying input to the common terminal. The output of NAND gate generates output as NOT A. For constructing AND gate, two NAND gates are required. The first NAND gate provides complement of A AND B., whereas the second NAND gate acts like an inverter. The double complement will cancel each other to provide AND gate.

Three NAND gates are required to construct the OR gate. First level two NAND gates will act like an inverter. These two inverters will complement the inputs. The third NAND gate will perform ANDing of the complemented inputs first and then inverts the result. This provides final output as logical OR functions.

NOR Gate as universal logic gate

Let us consider now NOR gate as a universal logic gate. Figure 10 shows the implementation of basic logic gates using only NOR gates.

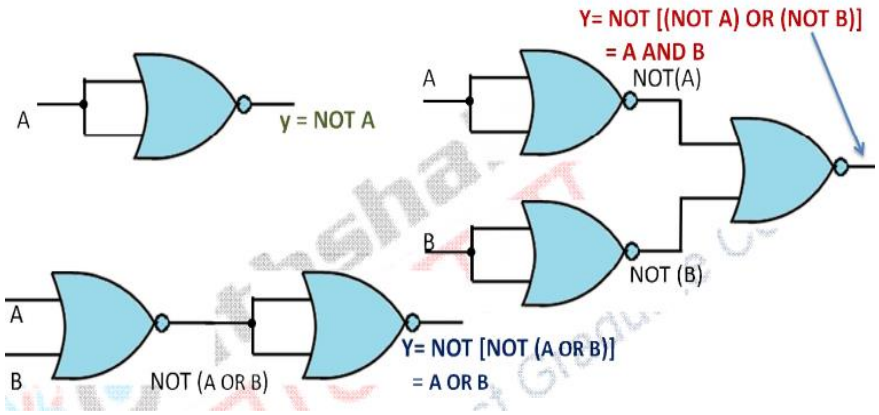


Figure 10: Construction of NOT, AND & OR gates using NOR gates

A NOR gate can be used to as NOT gate by connecting all its inputs together and applying input to the common terminal. The output of NOR gate generates output as NOT A i.e. complement of A. The construction of OR gate requires two NOR gates. The first NOR gate provides complement of A OR B. Whereas the second NOR gate acts like an inverter. The double complement will cancel each other to provide action of OR gate.

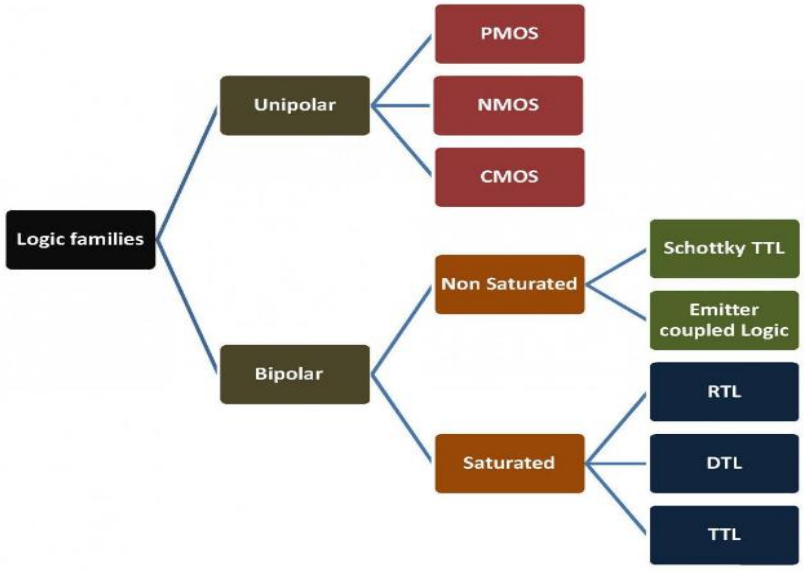
Three NOR gates are required to construct the AND gate.

In the first level two NOR will gates act like an inverter. These two inverters will complement the inputs. The third NOR gate will perform ORing of the complemented inputs first and then inverts the result. This provides final output as logical AND functions.

Realization of discreet logic gates (lecture class note)

Digital Ics and logic families

Most electronic systems which are responsible for modern advances are based on digital technology. All digital systems, computers and microprocessors are assembled from simple circuits called logic circuits. The basic building blocks of logic circuits are logic gates. And logic gates themselves are simple electronic circuits comprising of diodes, transistors and resistors.



Digital Integrated circuits

Digital Integrated circuits are produced using several different circuit configurations and production technologies. Each such approach is called a specific **logic family**. A **logic family** is a collection of different integrated circuit chip that have similar input, output, and internal circuit characteristics, but they perform different **logic gate** functions such as **AND**, **OR**, **NOT**, etc.

The idea is that different logic gate functions, when fabricated in the form of an integrated circuit with the same approach, or which belongs to the same logic family, will have identical electrical characteristics (electrically compatible with each other). These families may vary by speed, power consumption, cost, voltage and current levels.

In case of integrated circuits belonging to different logic families, digital system should ensure compatibility interfacing techniques. And that is the reason why we must be clear about different logic families and use the best combination of integrated circuits during the design of a digital system.

The characteristics which are bound to be identical and used to compare performance are:

1. Supply voltage range
2. Speed of response
3. Dissipation of power
4. Input and output logic levels
5. Current sinking capability
6. Current sourcing capability
7. Flexibility
8. Noise immunity
9. Fan-out

INTEGRATED CIRCUIT AND LOGIC FAMILY

Many logic families were produced as individual components, each containing one or a few related basic logical functions, which could be used as "building-blocks" to create systems or as so-called "glue" to interconnect more complex integrated circuits. A logic family may also refer to a set of techniques used to implement logic within very large scale integrated circuits such as central processors, memories, or other complex functions. Some such logic families use static techniques to minimize design complexity. Other logic families, such as domino logic, use clocked dynamic techniques to minimize size, power consumption and delay.

Before the widespread use of integrated circuits, various solid-state and vacuum tube logic systems were used for logic circuitry operations. But these were not standardized and interactive as the integrated circuit devices.

The most common logic family in modern semiconductor devices is Metal Oxide Semiconductor (MOS) logic, due to lower power consumption, small sized transistor, and high transistor density.

WHAT IS LOGIC GATES?

The electric circuits which perform logical operations are called Electric Gate. A Logic Gate is an electronic circuit that has two or more inputs but only one output. Logic follows well defined rules, producing predictable digital output from certain input.

Main Logic gates are **AND, OR, NOT, NAND, NOR and XOR**. Digital logic gates NAND and NOR are called universal logic gate because we can construct all other logic gates using NAND gate or NOR gate alone.

HOW THE LOGIC GATES FUNCTION?

In digital electronics, a gate is defined as an electronic circuit having only one output but two or more input. These electronic circuits consist of p-n junction diodes, transistors and resistors. Gates work with only two voltage levels at their input and output terminals with respect to ground or common, ie., the input and output voltages can have only two values. These values are typically chosen to be 0 Volt (zero volt) and some positive dc voltage value, say +5 Volt. For the 0 Volt logic level, it is customary to use the term that has two or more LOW (or the letter L) and the symbol 0 (zero). While referring to the positive voltage level, we use the term HIGH (or the letter H) or the number 1 (one). A signal appears at the output of a gate only for certain combinations of the input signals. This leads to a number of different types of gates. Gates enable us to perform several tasks which involve logical decision making, hence the nomenclature logic gates.

TYPES OF LOGIC FAMILY

The digital integrated circuits are designed using **bipolar devices** or **Metal Oxide Semiconductor (MOS)** or a combination of both. There are two kinds of semiconductor devices. The logic family which falls under the first kind **Bipolar logic family** and the other is **Unipolar logic family**.

BIPOLAR LOGIC FAMILY

There are two kinds of operations in bipolar integrated circuits: **Saturated Bipolar Logic family** and **Non-saturated Bipolar Logic family**.

Saturated Bipolar Logic Families are:

1. Diode logic (DL)
2. Resistor Transistor Logic (RTL)
3. Diode Transistor Logic (DTL)
4. Integrated Injection Logic (IIL or I2L)
5. Transistor Transistor Logic (TTL)

Non-saturated Bipolar Logic Families are:

1. Schottky TTL
2. Emitter Coupled Logic (ECL)

UNIPOLAR LOGIC FAMILY

Unipolar logic family consists of **Metal Oxide Semiconductor (MOS) logic families**. They are:

1. P-type MOS (PMOS) Logic
2. N-type MOS (NMOS) logic
3. Complementary MOS (CMOS) logic
4. Bipolar MOS (BiMOS) logic
5. Bipolar CMOS (BiCMOS) logic

CLASSIFICATION OF LOGIC FAMILY

Logic families are mainly classified as Bipolar Logic Families and Unipolar Logic Families.

1. **Bipolar Logic Families:** It mainly uses bipolar devices like diodes, transistors in addition to passive elements like resistors and capacitors. These are sub classified as saturated bipolar logic family and unsaturated bipolar logic family.
1. **Saturated Bipolar Logic Family:** In this family the transistors used in ICs are driven into saturation. For example:
    1. Transistor-Transistor Logic (TTL)
    2. Resistor-Transistor Logic (RTL)
    3. Direct Coupled Transistor Logic (DCTL)
    4. Diode Transistor Logic (DTL)
    5. High Threshold Logic(HTL)
    6. Integrated Injection Logic (IIL or I 2 L)
2. **Unsaturated bipolar logic family:** In this family the transistors used in IC is not driven into saturation. For example:
    1. Schottky TTL
    2. Emitter Coupled Logic(ECL)

2. **Unipolar Logic Families:** It mainly uses unipolar devices like MOSFETs in addition to passive elements like resistors and capacitors. These logic families have the advantages of high speed and lower power consumption than bipolar families. These are classified as:
    1. PMOS or P-Channel MOS Logic Family
    2. NMOS or N-Channel MOS Logic Family
    3. CMOS Logic Family

FEATURES OF LOGIC FAMILIES

1. TTL - Transistor-Transistor Logic: Standard logic family; used for the longest time.
2. ECL - Emitter Coupled Logic: Suitable for systems requiring high-speed operations.
3. MOS - Metal Oxide Semiconductor Logic: Suitable for systems with high component density.
4. CMOS - Complementary Metal Oxide Semiconductor Logic: Suitable for systems with low power consumption (VLSI circuits). Gradually becomes the dominant logic family.

Introduction to IC fabrication

Integrated Circuits are the set of electronic circuits mounted on the small electronic chip. Large numbers of devices, such as **MOSFETs** (Metal Oxide Semiconductor Field Effect Transistor), **transistors, microcontrollers, computer processors,** etc., are created with the help of the fabrication process. The most preferred materials in the semiconductor industry for fabrication are **silicon** and **germanium** because of their stable structure. Other materials preferred in fabrication industries are carbon steel, silver, aluminum, magnesium, copper, wood, thermoplastics, resins, etc. Let's first discuss fabrication and its requirements in the semiconductor field.

Fabrication is the process of constructing an industrial product. We can also define it as a set of methods to manufacture an electronic device or product. For example, **silicon semiconductor chips,** etc. In the case of metals, fabrication is a process used to convert the raw materials into the finished product. The smaller components manufactured by the fabrication process can be further used to make large electronic components.



The basic fabrication processes of the Integrated Circuits are as follows:

- Wafer Preparation
- Oxidation
- Diffusion
- Ion Implantation
- Chemical-Vapor Deposition
- Photolithography
- Metallization
- Packaging

Let's discuss the above processes in detail.

1, Wafer Preparation

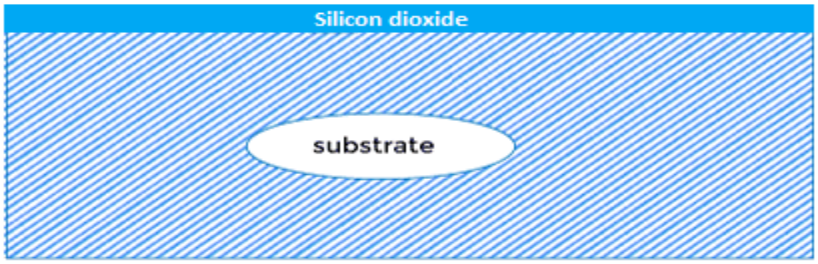
A wafer is a thin material used for making various **Integrated circuits** and **transistors.** Wafer acts as a base for such devices. The material of a wafer is the semiconductor, especially **crystalline silicon.** The silicon crystals used for the wafer manufacturing are highly pure. The process of extracting pure metal from the melt is known as a **boule.** The impurities are further added to the molten state of the material in a specific amount to make it n-type or p-type. The wafer preparation is the first step for IC fabrication.

It involves **cutting, shaping,** and **polishing** the wafer material to make it suitable for further fabrication. Some wafers are modified because of their sharp edges, irregular surface, and shape to convert them to the required wafer.

2, Oxidation

Oxidation is the process of adding **oxygen.** In a semiconductor, the oxygen and the silicon react to form silicon dioxide. The oxidation is carried out in furnaces at high temperatures upto 1250 degrees Celsius. Oxidation is classified as **wet** oxidation or **dry** oxidation. Both processes are widely used and have their own advantages and disadvantages. Wet oxidation is fast, while dry oxidation en has good electrical properties.

Wet oxidation is also known as **steam.** Both types of oxidation have excellent electrical insulation properties. The deposition of silicon dioxide on the silicon wafer protects from many impurities. The dopants can be applied only to areas not covered with the SiO2. An example of oxide layer on the substrate is shown below:



The oxidation process is one of the essential parts of the IC fabrication process and other VLSI (Very Large Scale Integration) devices.

3, Diffusion

Diffusion is a process of adding impurities atoms from a region with high concentration to a region of low concentration. The dopants or impurity atoms are added to the silicon (semiconductor material), which changes its resistivity. The process of diffusion is highly dependent on the temperature. It is carried out in high-temperature furnaces between 1000 degrees Celsius and 1200 degrees Celsius. The depth and width of the impurities depend on the temperature range and the timings. The high doping concentration improves the conductivity of a metal.The dopants can be of any state, solid, liquid, or gas. The preferred dopants are pentavalent impurities or n-type, such as antimony, phosphorous pentoxide, and arsine (gas). The trivalent impurities or p-type are **gallium, indium, boron,** etc.

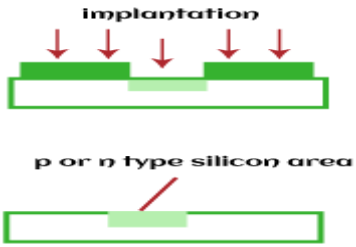
The diffusion cannot be used at the later stages. It is because the number of layers formed at the successive stages of the fabrication process might not be able to resist such high temperatures.

For example,

The boron or indium (trivalent impurities) diffusion on the n-type substrate forms a p-n junction. Diffusion is also a part of the process used to fill the lattice gap in the solids.

4, Ion Implantation

The ion implantation is a process of accelerating ions from the element to the solid target. The accelerated ions on the solid can change its properties. The ions are applied at a low temperature and high energy. The low temperature allows it to work at room temperature. But, the excess energy can also damage the crystal structure of the solid.



Ion implantation introduces impurity atoms in the crystal (semiconductor material). The accelerated atom gets embedded when it strikes the surface of the crystal. The energy and accelerating field voltage determine the depth of the penetration of the ions. The quantity of dopants can be controlled by adjusting the amount of ion concentration. Hence, ion implantation is more accurate due to the controlled current, voltage, and energy. The process is generally used when an accurate amount of impurity atoms is necessary for the device's operation.

5, Chemical-Vapor Deposition

CVD or Chemical Vapor deposition is used to produce **high-quality solid materials.** It produces thin films in the semiconductor industry. The process is carried out at a pressure below the atmospheric pressure, also known as **vacuum deposition.** The chemicals and vapors react to form solids on the surface of a substrate. The protective layers on the substrate, such as **silicon dioxide, polysilicon,** and **silicon nitride,** are deposited using the CVD process.

6, Photolithography

Photolithography is also known as **optical lithography,** using light to produce thin films. It is used to pattern thin films on the substrate, such as silicon wafers. It protects certain areas during successive fabrication processes (deposition, ion implantation, etc.). The types of light used to introduce mask patterns on the silicon wafer include **UV light, X-rays,** and **extreme UV** that emits at different frequencies. The most used type of light to produce the film is Ultra-Violet light.

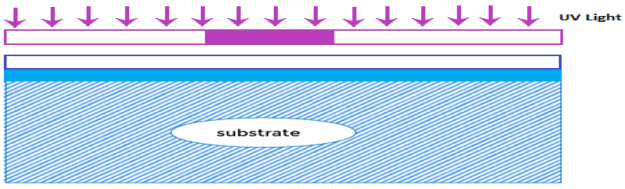
The silicon wafer or the substrate is first coated with a photoresist. On the photoresist layer, a mask pattern with the help of photolithography is applied to the silicon wafer. The exposed areas of the wafer become soft and can be removed. It creates a pattern on the wafer. The wavelength of each type of light determines the feature size impressed on the photoresist. Photolithography is somewhat similar to photography because the exposed area is patterned with help of the light. The incident light can be direct or projected through the lens.

UV or Ultra-Violet Light

UV light is exposed on the wafer with the help of a **mask.** It transfers the geometric design from the mask to the surface of the wafer coated with a photoresist. When the UV light falls on the exposed area on the substrate, the photoresist either breaks or hardens. The softened parts of the coating are further removed using the **specific solvent.**

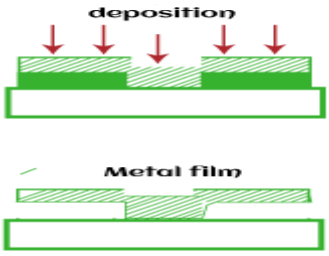
Photolithography is commonly used for the fabrication of microprocessors and solid-state devices. It can create a pattern of different sizes as small as **nanometers.** It is a quick, low-cost, and efficient method to create a substrate pattern in a single step. But, it is not ideal for irregular surfaces. In the case of complex IC circuits, the fabrication cycle can extend upto 50 times due to the microscopic structures.

The light falling on the substrate can be UV or X-rays, as shown below:

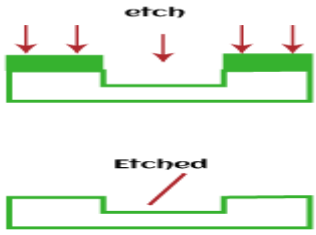


7, Metallization

Metallization is defined as the process of coating a metal layer on the metallic surface or non-metallic surface. The coating can be of **aluminum, zinc,** or **silver.** The metal coating in CMOS fabrication is aluminum, which protects the surface from external environmental factors, as dust, air, water, etc.



Metallization is also used to interconnect various components that form an Integrated circuit. The components can be resistors, capacitors, transistors, relays, etc. The metal layer is first deposited on the surface of the silicon wafer, as discussed above. After that, the required pattern or area for interconnected is etched, as shown below:



The metallization process uses chamber to apply metal layer. The wafer is placed inside the chamber, which coats the entire surface inside it. The thickness of the metal layer can vary depending on the requirements.

8, Packaging

Packaging is the last stage of the IC fabrication process. The finished silicon wafer or chips of various sizes with number of small components are tested electrically to recheck their working. The testing is performed using an automatic probing station. It is a cost effective testing machine that includes microwave and radio-frequency testing. The default circuits are separated from all the circuits. The good circuits are sent for packaging or headers.

Packaging also assembles the ICs with other devices. IC is a component of a device. The external connections with the IC are packed together to form a device. We can also say that packaging is a connection between the manufacturing and end use of a device. It makes a product suitable for its end use. The package is finally sealed with **epoxy** or **plastic** to prevent it from atmospheric dust. It is performed under vacuum or an inert atmosphere, which prevents the material from oxygen that may get trap in the packaging.

Conclusion on IC fabrication process

Let's conclude the above IC fabrication processes.

The IC fabrication starts with the wafer preparation, oxide layer deposition or a protective layer, introducing impurities into the wafer (n-type or p-type), implantation of ions to the surface of the crystal, CVD to produce thin films, photolithography (creating a mask pattern with the help of UV light), deposition of the metal for interconnections, and the packaging.